

Technical Documentation

Diesel Engine
12 V 183 TD13
Base: OM 444 LA

- EURO 1 -

Rail Traction

Operating Instructions
M015470/00E



Printed in Germany

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Preliminary Remarks/Important Information

Operation and maintenance are important factors in determining an engine's availability for operation and length of service life. These Operating Instructions are intended to provide clear answers to important questions regarding operation and maintenance.

We strongly recommend that the specified maintenance tasks be carried out carefully and on schedule. In order not to jeopardise warranty claims, conditions for engine operation which deviate from the normal operating conditions must be taken into consideration.

In event of questions concerning customer service, repairs or spare parts for the engine, please first contact the supplier of the equipment or component. Always state the complete engine no. given on the engine nameplate. If necessary, the supplier will use the MTU or MERCEDES-BENZ Service.

Furthermore, MTU's Product Support Service is available in an advisory capacity.

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Notes for the user

This Manual consists of 10 Sections, marked by tab-index sheets.

These Sections are listed on this page.

Each Section has its own detailed Table of Contents.

Using the Table of Contents to trace an entry

- ↓ Table of Contents (general)
- ↓ Contents
- ↓ Summary/Page

Using the Alphabetical Index to trace an entry

- ↓ Alphabetical keyword
- ↓ Heading/Page

Amendment Service

Please complete and return the Confirmation of Receipt card; you will then receive amendments to your manual.

1 Safety Instructions

Safety and accident prevention measures and instructions

2 Description

System and component descriptions

3 Operation

Initial operation, starting, monitoring and shutdown

4 Maintenance

Maintenance intervals, schedules and frequency charts, maintenance tasks

5 Cleaning, Preservation

Cleaning and preservations specifications

6 Practical Advice

Faults, causes and remedies, test and assembly work, injector replacement

7 Fluids and Lubricants

Safety instructions, diesel fuel, engine oils and coolants

8 Technical Data

Engine data, test values and settings, tightening torques

9 Keyword Index

Alphabetical index of key terms

10 Appendix

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1.1 General

The engine is a state-of-the-art product and conforms with all applicable specifications and regulations. Nevertheless, persons and property may be at risk in the event of

- incorrect use
- operation by unqualified personnel or personnel not trained by Mercedes-Benz or MTU
- incorrect alteration or conversion
- safety instructions being disregarded

1.2 Correct Use

The engine is intended solely for the use specified in the contract. Any other use is considered improper use. MTU and Mercedes-Benz will accept no liability for any resultant damage. The responsibility is borne by the user alone.

Liability for any damage resulting from this use is borne solely by the manufacturer of the finished machine/vehicle.

These Operating Instructions must be followed. The maintenance and care specifications must be followed.

1.3 Symbols

	Risk of injury!
	This symbol is used with all safety instructions concerning a hazard which, if disregarded, may result in immediate danger for human beings.

NOTE:

Safety instructions are given primarily in cases where, if disregarded, there is the risk of material, functional and environmental damage.

Instructions for environmental protection are also marked.

Other instructions provide useful information on engine operation and maintenance.

1.4 Personnel Requirements

Work on the engine must be carried out only by appropriately qualified personnel, or personnel trained by MTU or Mercedes-Benz, including personnel of workshops authorised by MTU or Mercedes-Benz.

Responsibilities of the operating, maintenance and repair personnel must be specified.

The specified legal minimum age must be respected.

1.5 Conversion and Alteration of the Engine

Unauthorised alterations to the engine may affect its operation and safety. No liability for resultant damage is accepted.

Similarly, manipulation of the injection and governor systems may affect the performance and exhaust emissions of the engine. Factory settings and fulfilment of statutory environmental conditions are then no longer guaranteed.

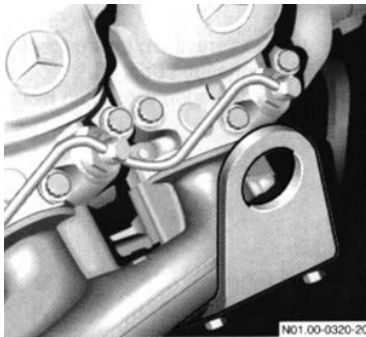
1.6 Transportation

Use only the transport equipment approved by MTU or Mercedes-Benz. Follow lifting specifications.

NOTE:

The engine must only be transported in installation position.

Use only the engine's own lifting fixtures.
(See engine installation drawing)..



1.7 Installation

The engine is intended solely for the use specified in the contract. See Sections "Proper Use" (Page 5) and "Conversion and Alteration of Engine" (Page 6).

These Operating Instructions contain data for installation in the "Technical Data" Section. In event of queries, please contact your MTU or Mercedes-Benz Service.

1.8 Organisational Measures

These operating instructions must be issued to the personnel involved with operating or working on the engine and kept at the engine's site of operation, where they should be accessible at all times.

The personnel must be instructed on engine operation by means of the Operation Manual, and in particular the safety instructions must be explained. This is especially important for personnel who work on the engine only on an occasional basis.

In addition to the operating instructions, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed.

1.9 Safety measures on engines with electronic engine management ECS-183/1

All adjustable parameters (limits and trip points) for engine control and monitoring must be coordinated with the responsible project department SVR at MTU Friedrichshafen.

It is essential to carry out the following safety measures to prevent damage to the engine, components and wiring, as well as injury to personnel.

NOTE:

- Start engine only when batteries are securely connected.
- Do not disconnect batteries when engine is running.
- Start engine only when speed transmitters are installed and connected.
- Do not use boost chargers to start engine. Start assistance only with separate batteries.
- To boost charge the batteries, the battery terminals must be removed. See operating instructions of boost chargers.
- For electro-welding work, the batteries must be disconnected and the two cables (+ and -) securely connected together. Separate plug-in connections of control units.
- The plug-in connections of the engine governor must only be disconnected and connected with the electrical system switched off.
- Incorrect polarity of the engine governor supply voltage (e.g. incorrect polarity of the batteries) can result in destruction of the engine governor.

- It is essential to tighten the union nuts of the electric connectors (controlling mechanism for injection pump sliding-sleeve, temperature and pressure sensors) to specified torque - see "Technical Data".
- If the two speed transmitters are interchanged in the engine control housing, this may result in damage from overspeeding during the starting sequence.

1.10 MTU and Mercedes-Benz Original Components

The original components are subjected to very stringent quality checks and guarantee maximum function, safety and value retention. Each component is specially designed, produced, selected and approved for MTU and Mercedes-Benz.

We must therefore reject any responsibility for damage resulting from the use of components and accessories which do not satisfy these conditions.

NOTE:

For economical repair in the course of recycling, MTU and Mercedes-Benz exchange units and components are offered, which come with the same quality standards and guarantee as new components.

In the Federal Republic of Germany and some other countries, certain components, for example safety components, are permitted for installation and conversion only if the applicable legal requirements are satisfied. This requirement is fulfilled in every case by MTU and Mercedes-Benz original components.

If other components which have not been tested and approved by MTU or Mercedes-Benz are used - even if, in individual cases, they have been accepted or officially authorised - this cannot be assessed or guaranteed, despite continuous market observation.

Warranty claims may therefore be restricted.

When ordering original components, and in event of technical queries, always state the engine number with type test designation.

NOTE:

In order to determine exactly the design status, the complete engine number and engine data card (see Page 14) are always required.

1.11 Safety and Emergency Propulsion Program (See also signal description of electronic engine management ECS-183/1 in "Appendix").

The engines have an electronics system which monitors both the engine and itself (self-diagnosis).

As soon as the system detects a fault, one of the following measures is automatically initiated, according to evaluation of the fault:

- Display of faults during operation by means of a fault code display installed in connection interface 1
- Detected faults, even temporary ones, are stored by the system in decimal code and issued in binary code.
- Changeover to suitable substitute function for further, though unrestricted engine operation (e.g. reduced speed).
- Immediate engine shutdown, if necessary for safety reasons. According to fault evaluation, the engine is shut down by reducing delivery to zero, and emergency shutdown is initiated.

NOTE:

The fault must be rectified immediately by the responsible MTU or Mercedes-Benz Service.

1.12 Automatic Emergency Shutdown

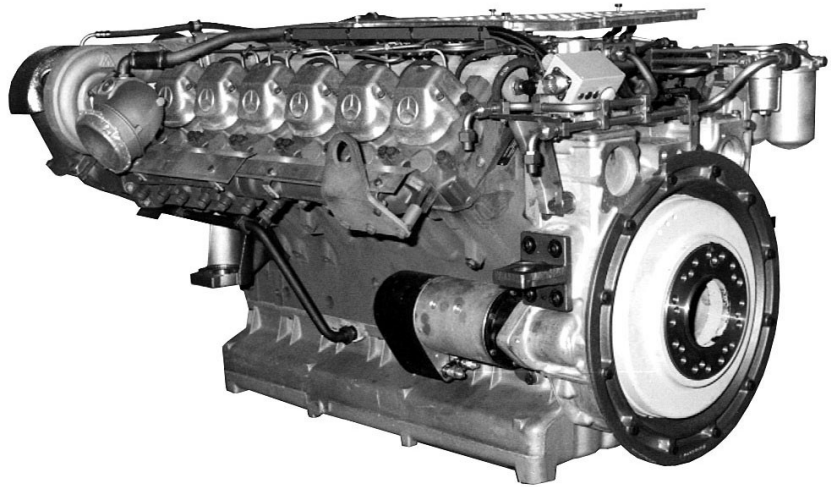
Any required emergency engine shutdown is effected by the electronic engine management system ECS-183/1 via the engine governor.

The governor moves the fuel injection pump control rack to zero delivery and the engine shuts down.

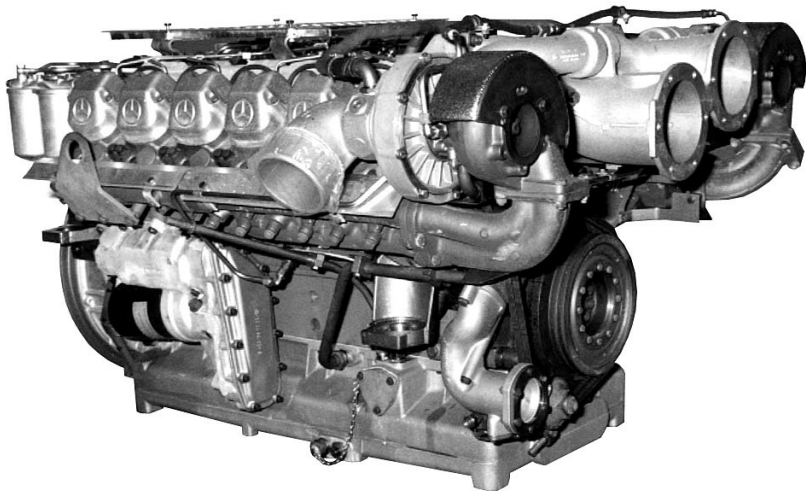
The engine can be restarted after an "Emergency shutdown". If the cause for the shutdown is not rectified, another "Emergency shutdown" is initiated. If an "Emergency shutdown" is initiated during a start attempt, the starter will turn but the engine will receive no fuel and will not start.

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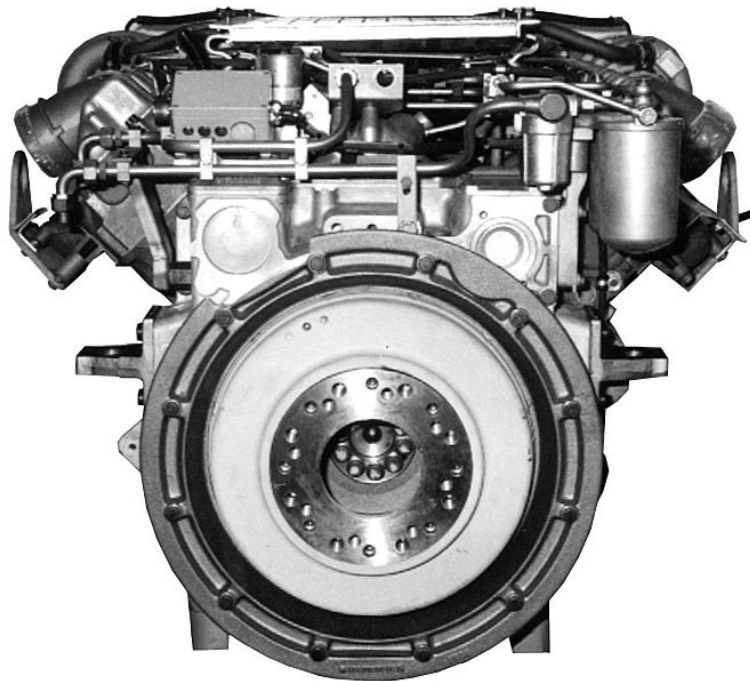
**2.4 Overview of 12 V 183 TD13
(View of driving end, left engine side)**



**2.4.a Overview of 12 V 183 TD13
(View of free end, right engine side)**



**2.4.b Overview of 12 V 183 TD13
(View of driving end)**



**2.4.c Overview of 12 V 183 TD13
(View of free end)**

